

LARRY HUI

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EDUCATION

M.S. in Robotics, University of Pennsylvania (GRASP Lab)

Expected May 2028

Coursework: World Models, Control and Optimization for Robotics, Machine Perception, Distributed Robotics

B.S. in Mechanical Engineering (EECS focus), University of California, Berkeley

May 2026

GPA: 3.59/4.0 (last two years: 3.91/4.0) | 2× Dean's Honors List (Top 10%)

Coursework: Robotics, Bipedal Robot Control, Convex Optimization, Machine Learning, Feedback Controls, Mechatronics, Deep RL for Controls, Embedded Systems, Continuum Mechanics, FEA

EXPERIENCE

Mechanical Engineering Intern, [Outer Rim Exploration Inc.](#), Tokyo, Japan

Jul 2024 – Dec 2024

- Designed the CAD assembly for a drone-mounted muon hodoscope, integrating PCB layouts and validation hardware.
- Designed a geared, solenoid-locked rotation stage that sets the detector's zenith angle to degree-level accuracy.
- Reduced mass by 10% and increased stiffness via geometric optimization, verified with custom MATLAB FEA solver.
- Cut prototype cost by 40% with COTS parts, helping secure \$200K in seed funding and a first pilot contract.

Undergraduate Researcher, [Gu Research Group](#), UC Berkeley

Dec. 2024 – May 2026

- Built PINN and physics-informed MeshGraphNet solvers for linear-elasticity PDEs in MATLAB and PyTorch for a MathWorks-sponsored project, using ensembles, a **192-mesh** training pool with metadata, and cosine annealing.
- Benchmarked PINN ensembles against FEA across 3 geometries (Timoshenko beam, plate with central hole, clamped plate), 3 BC types, and 4 loadings, achieving **<1.3%** displacement error over **15k+ collocation points**.
- Built a differentiable GNN solver for fluid–solid interaction with normalized input/output features and LayerNorm; a fluid-only warm start cut flap-displacement error from **7–32%** to **<5.2%** on unseen geometries.

Lead Teaching Assistant, [Mechanical Engineering](#), UC Berkeley

May 2025 – May 2026

- Led ME 103 Measurements (180+ students) and ME 132 Feedback Control (80+), coordinating 8 TAs and readers.
- Created labs & exams on SP, controller design with Root Locus, FR, and state-space in LabVIEW, MATLAB, Simulink.

PROJECTS

Cable-Actuated Air Hockey Robot

Jan. 2026 – May 2026

- Co-designed a 4-BLDC-motor cable-driven robot on an 80/20 frame, minimizing moving inertia for fast mallet motion.
- Developed a RT-MPC planner with quintic-spline trajectories & PPO policy scoring 83% of shots vs. 29% for Q-learning.
- Tracked the puck to **<0.1 mm** by fusing homography- and parallax-corrected OpenCV detections in an EKF.
- Derived closed-form IK and least-squares FK, streaming closed-loop commands to moteus motor controllers over CAN.

Autonomous Cable Insertion with UR7e Arm

Aug. 2025 – Dec. 2025

- Built a ROS2 IK/CV-control stack that localizes cables and ports with RealSense depth and detects USB-C, achieving mm-level insertion accuracy across randomized trials.
- Designed a custom end-effector and force limited-compliant insertion with retry/re-grasp recovery from misalignment.

Pixel Mirror: RT LED Display with Virtual Gloving

Jan. 2026 – May 2026

- Achieved <70 ms end-to-end latency at 30 fps by porting a YOLOv8-seg silhouette pipeline to native C++; streamed frames over a 1 Mbaud serial link to ESP32 firmware that drives LED board via I2S DMA with minimal CPU overhead.

RL for Humanoid Locomotion on Rough & Transient Terrain

Aug. 2025 – Dec. 2025

- Trained a Unitree G1 walking policy in IsaacLab via PPO (2,048 parallel envs) on a rough-terrain curriculum.
- Implemented BeamDojo's double-critic PPO from scratch; built moving stepping-stone terrain in MuJoCo MJX.

PUBLICATIONS

1. **L. Hui**, J. Kam, W. Su, and J. Zhou, "Hamilton–Jacobi Reachability for Spacecraft Collision Avoidance," *2026 IEEE International Conference on Control & Automation (ICCA)*, 2026.

SKILLS AND INTERESTS

CAD & Simulation

Languages/Frameworks

Libraries/Tools/OS

Hardware/Embedded:

Languages/Interests

SolidWorks, Fusion360, OnShape, ANSYS, ABAQUS, Simulink, Gmsh, GD&T, DFM

Python, ROS2, PyTorch, NVIDIA Isaac, C++, MATLAB, Java, Git, LabVIEW

NumPy, SciPy, Pandas, MuJoCo, CVX, Linux (Ubuntu), macOS, FreeRTOS

ESP32, Jetson Nano, CAN, UART, I2S/DMA, Machining, 3D Printing, Instron

English, Mandarin, French (B2); Private Pilot (FAA); ARCT piano diploma